DEPARTMENT OF MATHEMATICS
UNIVERSITY COLLEGE OF ENGINEERING (A)
OSMANIA UNIVERSITY, HYDERABAD, TELANGANA-07
B.E (CSE) – IV–SEMESTER– CLASS TEST –I
ENGINEERING MATHEMATICS–IV
(PROBABILITY &STATISTICS)

Academic Year: 2025-2026
Date: 23-03-2026

Max Marks: 30
Duration: 1hr 30 min

		M	BT	CO
	PART A (5x2=10 M) Answer All Questions			
1. a	Define the types of kurtausis for data.	2	1	1
b	Write normal equations to the cubic polynomial	2	2	4
c	Calculate variance of Poisson distribution.	2	3	3
d	Two players A and B play tennis games their chances of winning a game are in the ratio 3:2 resp. find A's chance of winning at least two games out of the five games played.	2	4	3
f	Find the rank correlation for following date of ranks in two subjects	2	3	4

Maths	9	6	4	3	2	5	1	8	7
Stats	5	8	4	2	1	3	6	7	9

	PART B (2x10= 20 M) Answer any TWO Questions								
2. a	Calculate the first four Moments of the following distribution about the Mean and comment upon the skewnes and kurtosis of the following marks distribution of the students.						5	4	1

Marks	5	10	15	20	25	30
No.students	2	5	8	7	5	3

b	Compute the Measures of central tendency for the following distribution of the scores(s) of 50 college students						5	4	1

s	140-150	150-160	160-170	170-180	180-190	190-200
f	4	6	10	18	9	3

3. a	Obtain the equation of the two lines of regression for the data given below						5	3	4

x	2	3	5	7	11	13
y	4.194	4.291	4.485	4.678	5.067	5.261

b	Fit a curve of the form $y = a.b^x$ to the following data using method of Least square.						5	3	4

x	2	3	4	5	6
y	144	172.8	207.4	248.8	298.5

4. a	Calculate varience and moment generating function of Binomial distribtion.						5	3	3
b	The following mistakes per page where obsrved in a book. Fit a Poisson distributionand estimate the expected frequencies.						5	4	3

No. of mistakes	0	1	2	3	4
No. of pages	211	90	19	5	0